Element A

Our problem is that the filaments for the 3D printers are very disorganized and messy. This problem affects the people who interact with the printer daily, like Mrs. Barletta, Matt, and Mr. Thib. This can also affect anyone who comes into the room and sees it. When someone has a 3D printer with multiple rolls of filament they have nowhere to store it. For people who own a 3D printer or have items for a 3D printer, this may be a bigger issue depending on what they have. This problem occurs in a classroom where the printers are taking up room for other items. This occurs whenever students have to put away the filament but they have nowhere to put it. This problem happens very constantly because there's almost nowhere to put the filament. There is a lot of filament lying on the counters with nowhere to go, so with these new filament holders, there will be a place to put the filament. Stakeholders for the printers are Mr Thib, Mrs. Barletta, and Matt. These people all use the printers. Mrs. Barletta is the consumer and uses the prints that are made. Mr. Thib and Matt can repair and add on to the printers. Matt finds designs that we can print and he can also create designs. Mr. Thib fixes the printers if they are beyond Matt's capabilities.

Element B

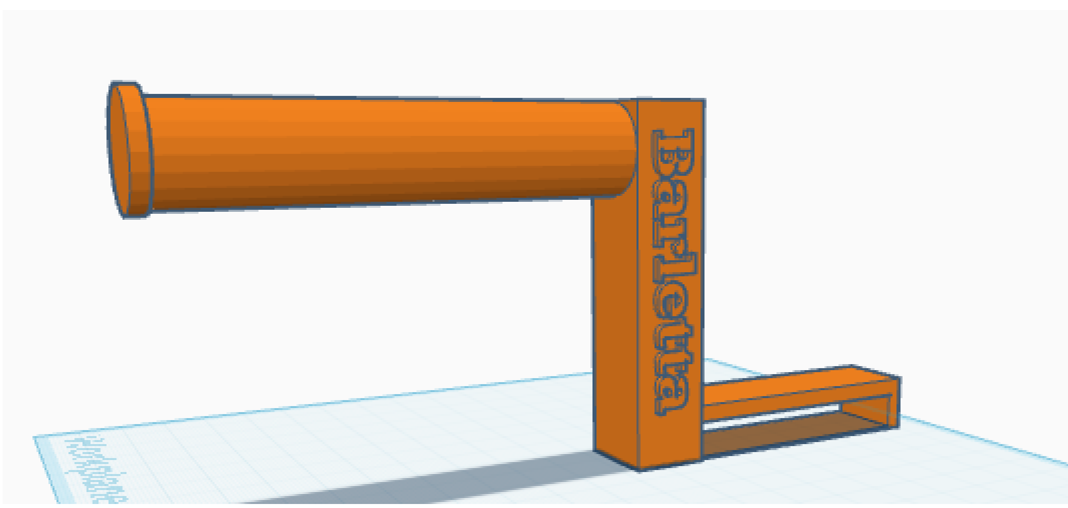
Matt made multiple failed attempts at building a filament holder for the 3D printer. The problem is that TinkerCAD does not transfer the correct measurements to Cura. In his first attempt, he designed a simple holder, but it snapped during the print process, which led him to rethink his approach. The material he chose for the print didn't have the right strength for the task. In attempt two, he resized the design, only to find that it was too short and didn't fit on the printer at all. It was frustrating to see the time wasted on what he thought would be a quick fix.

His third attempt involved a more complex tree design, but it turned out to be missing several crucial parts. He felt optimistic during the setup, but just as it was about to be finalized, he realized it barely fit on the printer and wouldn’t be able to support the filament properly.

Element C

We are making a filament holder that holds filament

1. High priority filament materials.
2. High priority measurements.
3. High priority matt.
4. High priority cleb being not sick.
5. High priority having the print be successful.
6. Low priority color.
7. Low priority weight.



Element D

Elephant E

STEM stands for science, technology, engineering, and math. First we used the scientific method to decide what we needed to do to make a filament holder, we first made a holder without measurements then fixed the measurements after.We used tinker cad which allows us to design the filament holder and fix any mistakes. We used the 3d printer as engineering and was able to print a successful prototype . We used math by measuring and the measurement of the holder then fixed any mistakes we made. We had our project checked by two experts Matt and Mr Thib that examined our work. They have experiences with 3d printing which made them good for the job.